FIBRES OF THE HENSELIZATION OF THE NORMALIZATION AND THE MINIMAL PRIMES OVER A LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span includes the two-arrow diagram Spec(A') <- Spec((A')^h) -> Spec(A^h), delivered verbatim. The ten references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Henselization of a local ring, its flatness and the identity (A/I)^h = A^h/IA^h. used in the counted proof span. Is the base of the reduction to the reduced case and of the fibre computations.
- The integral closure in the total ring of fractions as a finite product of normal domains, and normality of a filtered colimit of etale algebras. used in the counted proof span. Gives uniqueness of the minimal prime inside each maximal ideal of (A')^h.
- Henselian pairs and connectedness of the fibre of the maximal ideal. used in the counted proof span. Supplies the converse uniqueness of the maximal ideal containing a minimal prime.

Source reference. Original lemma 030C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3943–3943. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 037D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4196–4196. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 654–654. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00GU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 620–620. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 618–618. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 033C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4051–4051. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7594–7594. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04MG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6042–6042. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 2753: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2753–2753. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 2948: sources/stacks-more-algebra/pinned-original-source.tex, source lines 2948–2948. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a local ring. Assume A has finitely many minimal prime ideals. Let A' be the integral closure of A in the total ring of fractions of A_{red}. Let A^h be the henselization of A. Consider the maps*

$$\mathrm{Spec}(A') \leftarrow \mathrm{Spec}((A')^h) \to \mathrm{Spec}(A^h)$$

where $(A')^h = A' \otimes_A A^h$. Then

1. *the left arrow is bijective on maximal ideals,*
2. *the right arrow is bijective on minimal primes,*
3. *every minimal prime of $(A')^h$ is contained in a unique maximal ideal and every maximal ideal contains exactly one minimal prime.*

Proof. Let $I \subset A$ be the ideal of nilpotents. We have $(A/I)^h = A^h/IA^h$ by (Algebra, Lemma 05WQ). The spectra of A, A^h, A', and $(A')^h$ are the same as the spectra of A/I, A^h/IA^h, A', and $(A')^h = A' \otimes_{A/I} A^h/IA^h$. Thus we may replace A by $A_{red} = A/I$ and assume A is reduced. Then $A \subset A'$ which we will use below without further mention.

Proof of (1). As A' is integral over A we see that $(A')^h$ is integral over A^h. By going up (Algebra, Lemma 00GU) every maximal ideal of A', resp. $(A')^h$ lies over the maximal ideal $\mathfrak{m}$, resp. $\mathfrak{m}^h$ of A, resp. A^h. Thus (1) follows from the isomorphism

$$(A')^h \otimes_{A^h} \kappa^h = A' \otimes_A A^h \otimes_{A^h} \kappa^h = A' \otimes_A \kappa$$

because the residue field extension κ^h/κ induced by $A \to A^h$ is trivial. We will use below that the displayed ring is integral over a field hence spectrum of this ring is a profinite space, see Algebra, Lemmas 00GS and 04MG.

Proof of (3). The ring A' is a normal ring and in fact a finite product of normal domains, see Algebra, Lemma 030C. Since A^h is a filtered colimit of étale A-algebras, $(A')^h$ is filtered colimit of étale A'-algebras hence $(A')^h$ is a normal ring by Algebra, Lemmas 033C and 037D. Thus every local ring of $(A')^h$ is a normal domain and we see that every maximal ideal contains a unique minimal prime. By Lemma 2753 applied to $A^h \to (A')^h$ we see that $((A')^h, \mathfrak{m}(A')^h)$ is a henselian pair. If $\mathfrak{q} \subset (A')^h$ is a minimal prime (or any prime), then the intersection of $V(\mathfrak{q})$ with $V(\mathfrak{m}(A')^h)$ is connected by Lemma 2948 Since $V(\mathfrak{m}(A')^h) = \mathrm{Spec}((A')^h \otimes \kappa^h)$ is a profinite space by we see there is a unique maximal ideal containing $\mathfrak{q}$.

Proof of (2). The minimal primes of A' are exactly the primes lying over a minimal prime of A (by construction). Since $A' \to (A')^h$ is flat by going down (Algebra, Lemma 00HS) every minimal prime of $(A')^h$ lies over a minimal prime of A'. Conversely, any prime of $(A')^h$ lying over a minimal prime of A' is minimal because $(A')^h$ is a filtered colimit of étale hence quasi-finite algebras over A' (small detail omitted). We conclude that the minimal primes of $(A')^h$ are exactly the primes which lie over a minimal prime of A. Similarly, the minimal primes of A^h are exactly the primes lying over minimal primes of A. By construction we have $A' \otimes_A Q(A) = Q(A)$ where $Q(A)$ is the total fraction ring of our reduced local ring A. Of course $Q(A)$ is the finite product of residue fields of the minimal primes of A. It follows that

$$(A')^h \otimes_A Q(A) = A^h \otimes_A A' \otimes_A Q(A) = A^h \otimes_A Q(A)$$

Our discussion above shows the spectrum of the ring on the left is the set of minimal primes of $(A')^h$ and the spectrum of the ring on the right is the is the set of minimal primes of A^h. This finishes the proof. □